Questions for the tutor class:

1, 3, 7, 8, 9, 10, 11, 14, 15, 16, 17, 19, 20, 22, 24, 26

1. A man walks 8 m to the east followed by another 6 m to the south. What is his eastward displacement?
   A. 2 m
   B. 6 m
   C. 8 m
   D. 14 m

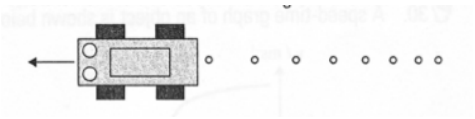
2. A car accelerates at a rate of 5m/s^2 from rest for 10s. What is its speed at the end of 10s?
   A. 0 m/s
   B. 2 m/s
   C. 5 m/s
   D. 50 m/s

3. A racing car accelerates uniformly along a straight road. Marker posts are placed at 100 m apart along the road. The car has a speed of 10 m/s when it passes one post, and 20 m/s when it passes the next. What is the acceleration of the car?
   A. 0.67 m s^{-2}
   B. 1.5 m s^{-2}
   C. 2.5 m s^{-2}
   D. 6.0 m s^{-2}

4. A metal ball hits a sand bed one second after it is released from rest, and makes an impression 8.0 mm deep in the sand. Air resistance may be taken as negligible. What is the average deceleration of the ball on hitting the sand?
   A. 6.0 x 10^2 m s^{-2}
   B. 1.2 x 10^3 m s^{-2}
   C. 6.0 x 10^3 **m** s^{-2}
   D. 1.2 x 10^4 m s^{-2}

5. A car decelerates at a rate of 5 m/s^2 from an initial speed of 40 m/s. What is its speed after 6 s?
   A. 8 m/s
   B. 10 m/s
   C. 29 m/s
   D. 35 m/s

6. A car has a hole in the petrol tank and petrol is dripping off the tank at a constant interval.



Which of the following statement is true about the diagram shown

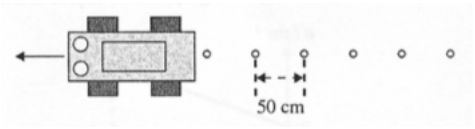
A. The car is moving at a constant speed

B. The car is accelerating

C. The car is decelerating

D. The car is moving at an increasing deceleration

7. A car has a hole in the petrol tank and petrol is dripping off the tank at a constant interval. Given that the car is travelling at a constant speed of 2 m/s and the gap between two drops of petrol along the road is 50 cm, calculate the frequency of the drip of the petrol.



A. 0.25 Hz

B. 4 Hz

C. 8 Hz

D. 25 Hz

8. For a particle accelerating along a straight line, what is the definition of its velocity?

A. Velocity is displacement divided by time

B. Velocity is distance divided by time

C. Velocity is the rate of change of displacement

D. Velocity is the rate of change of distance

9. Which of the following gives the acceleration of a body?

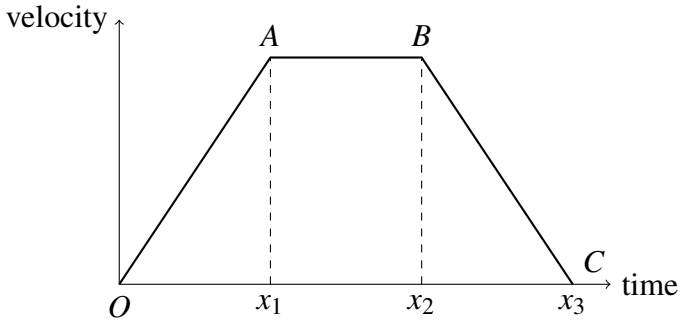
A. the area under its displacement - time graph

B. the area under its velocity - time graph

C. the gradient of its displacement - time graph

D. the gradient of its velocity - time graph

10. The figure denotes velocity−time graph. Displacement is equal to:



A. $y(x_1+x_2+x_3)$

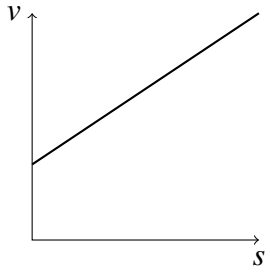
B. $y(x_3-x_1)$

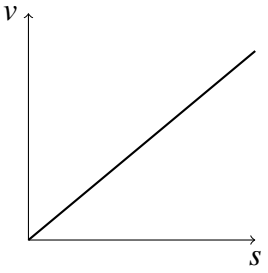
C. $y(x_3-x_2-x_1$

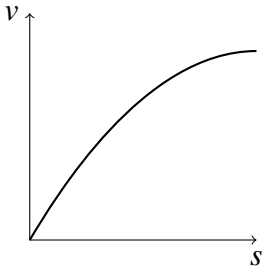
D. $\frac{y}{2}(x_3+x_2-x_1)$

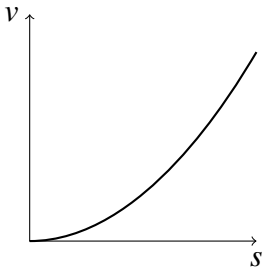
Solution: The area enclosed between velocity−time curve and the time axis gives displacement. $OABC$ is a trapezium. Side $OC = x_3$, side $AB = (x_2-x_1)$, height $=y$; Area = $\frac{1}{2}$∗height∗(sum of parallel sides) $= \frac{1}{2}y(x_3+x_2-x_1)$; Displacement = $\frac{1}{2}y(x_3+x_2-x_1)$

11. A body starting from rest moves along a straight line with a constant acceleration. The variation of speed v with distance s is represented by the graph?

A. 

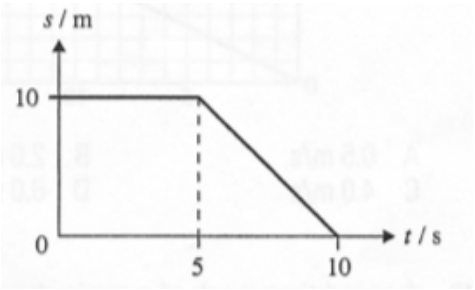
B. 

C. 

D. 

Solution: Use the equation $v^2 - v_0^2 = 2aS$, with $v_0 = 0$, $v^2 = 2aS$. This is an equation to parabola. It is symmetric about the axis of distance S. (C) denotes the graph

12. Which of the following best describes the displacement-time graph of an object?



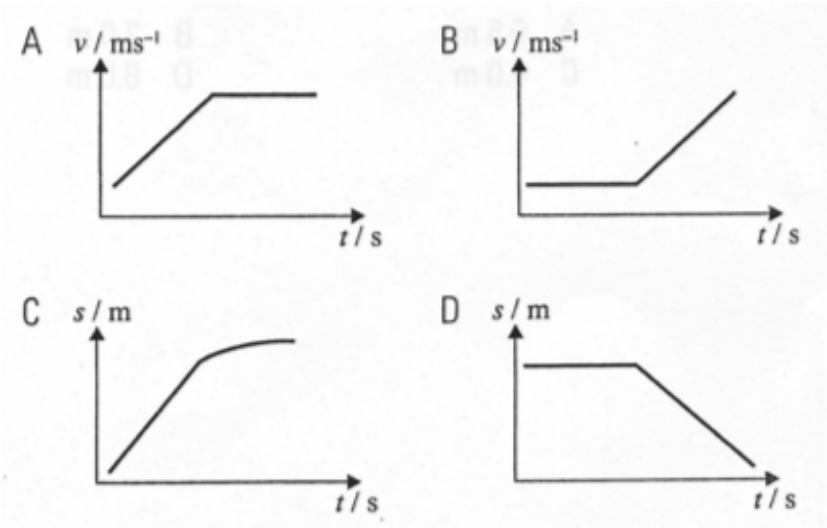
A. The object stops for 5 seconds before moving backward at a constant speed.

B. The object stops for 5 seconds before moving forward at a constant speed.

C. The object travels at a constant speed for 5 seconds before slowing down to rest.

D. The object travels along a level road for 5 seconds before going down a slope.

13. Which of the following diagram shows that an object is moving at a constant velocity before slowing down?



Solution: C

14. A robot shoots a ball vertically upwards at an initial speed of 30 m/s. What is the distance travelled by the ball 5 s after the shoot? ($g = 10m/s^2$)

    A. 25 m

    B. 45 m

    C. 65 m

    D. 75 m

15. A runner accelerates constantly at 2m/s^2 for the first 4 seconds in a 100m race. He then maintains the speed to complete the race. Calculate the time he takes to complete the 100m race.

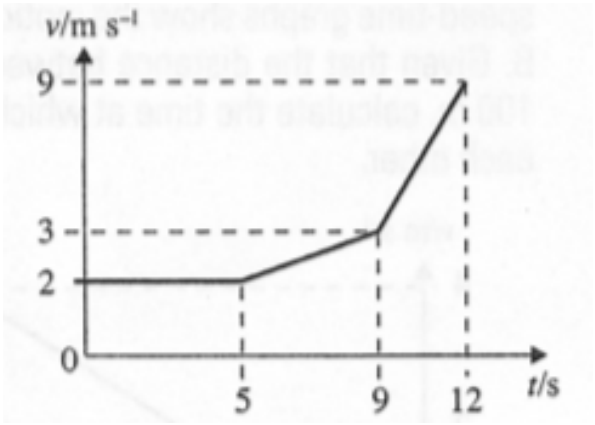
    A. 10.0 s

    B. 12.5 s

    C. 13.5 s

    D. 14.5 s

16. The speed-time graph of a motorcycle is shown below. What was the average speed over the 12 s?



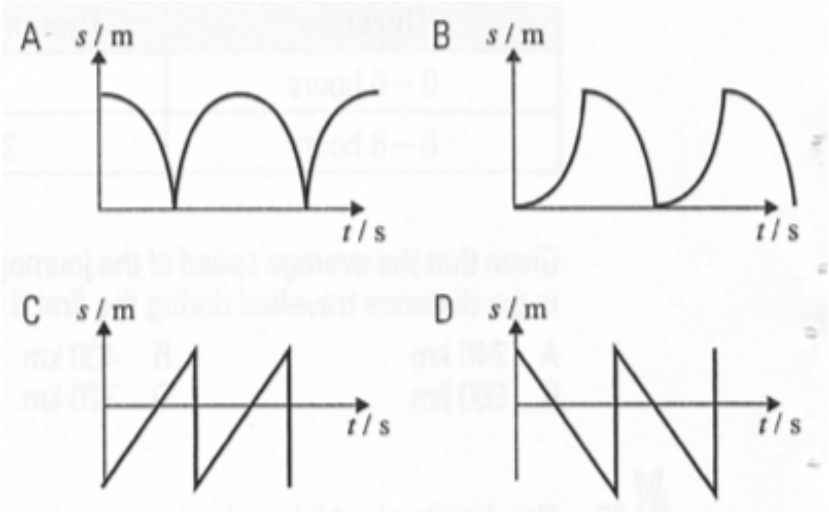
    A. 0.38 m/s

    B. 0.58 m/s

    C. 0.75 m/s

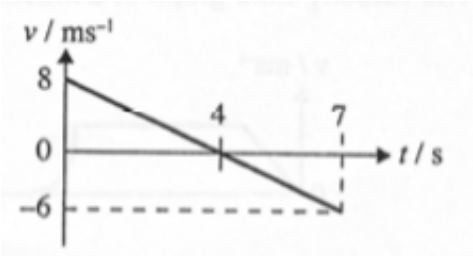
    D. 3.17 m/s

17. Which of the following graph is the displacement-time of a bouncing ball?



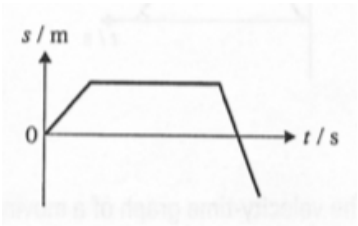
Solution: A

18. The velocity-time graph below shows the motion of an object. Calculate the distance travelled (S) and displacement (x) of the object.

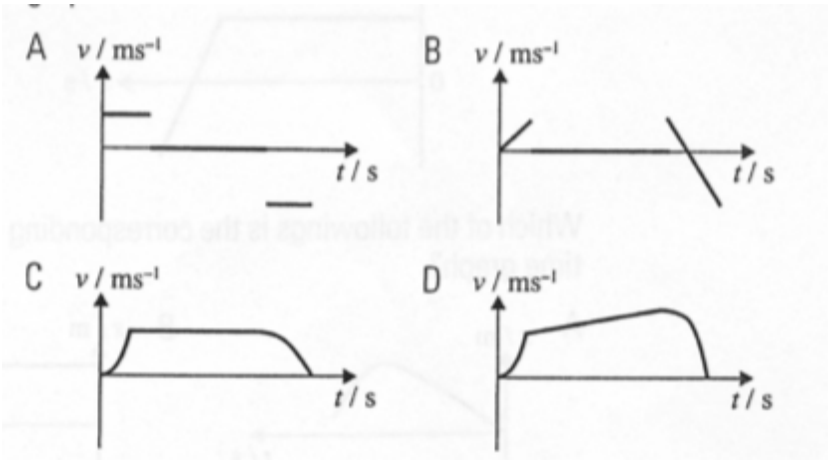


A. S = 49 m and x = 9 m
B. S = 16 m and x = 9 m
C. S = 49 m and x = 7 m
D. S = 25 m and x = 7 m

19. The displacement-time graph of a moving object is shown below:

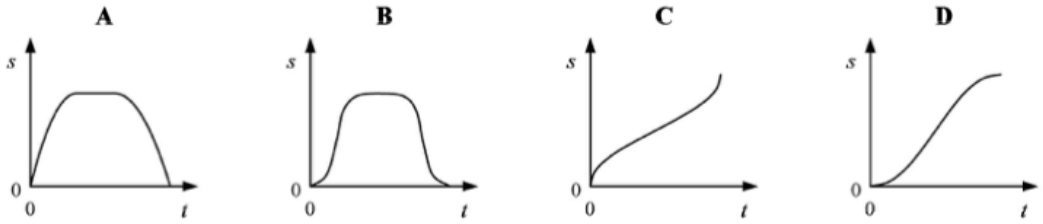


Which of the following is the corresponding velocity-time graph ?



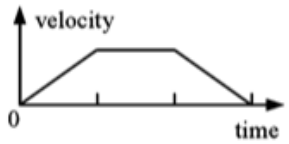
Solution: A

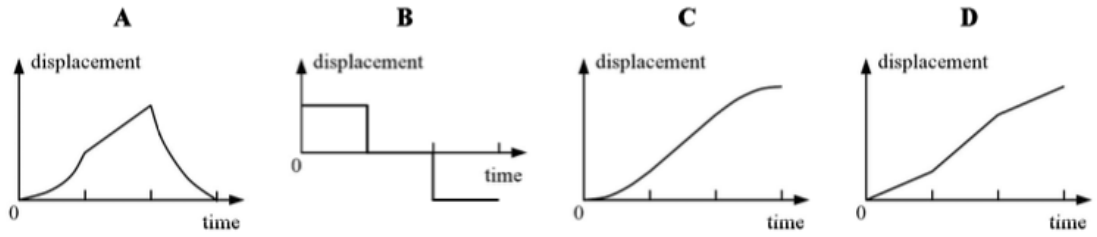
20. After accelerating down a hill, a cyclist moves at constant speed and then decelerates as he climes up another hill. Which graph show the variation of distance s traveled by the cyclist with time t?



Solution: D

21. The given graph shows the variation with time of the velocity of a moving body. Which graph best represents the corresponding variation of displacement with time?





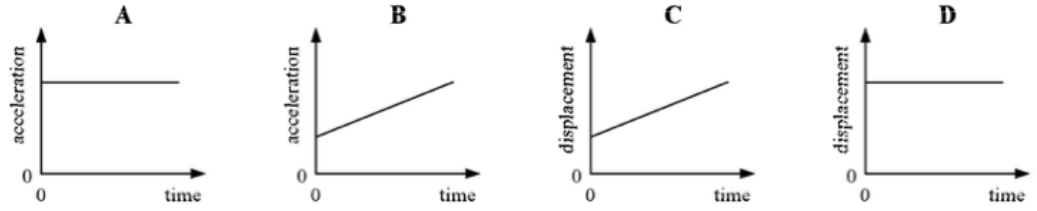
Solution: C

22. The average speeds of a car, accelerating uniformly through three gear changes, are as shown. What is the overall average speed of the car?

$20\,\text{m s}^{-1}$ for 2.0 s
$40\,\text{m s}^{-1}$ for 2.0 s
$60\,\text{m s}^{-1}$ for 6.0 s

A. 12 m/s
B. 13.3 m/s
C. 40 m/s
D. 48 m/s

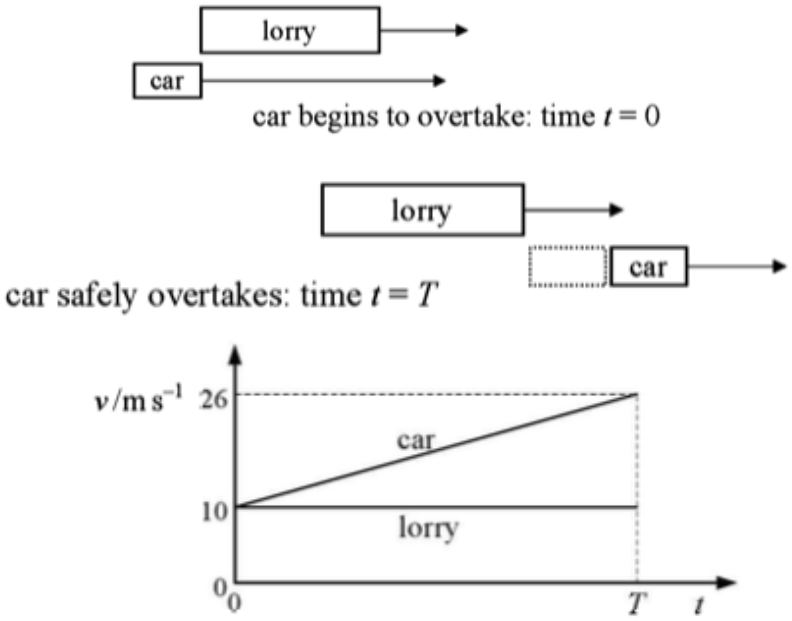
23. A car is travelling with uniformly increasing speed along a straight road. Which graph best represents the motion of the car?



Solution: A

24. The minimum time T required for a car to safely overtake a lorry is illustrated in the given diagram. It is measured from the time the front of the car is level with the rear of the lorry, until the rear of the passing car is a full car-length ahead of the front of the lorry.

The lorry may be taken to be 17.0 m long and the car 3.5 m long. The given graph shows how the speeds v of of the car and the lorry vary with time t.
what is the value of T?



A. 0.86 s

B. 1.2 s

C. 2.6 s

D. 3.0 s

25. A car starts from rest and accelerates uniformly. After 5.0 s, its speed is 20.0 m/s.

(a) What is the acceleration of the car?

(b) What is its speed after 2.0 s?

(c) What is the distance travelled in the first 2.0 s?

Solution:

(a) 4.0 m/s^2

(b) 8.0 m/s

(c) 8.0 m

26. A bus was travelling at 20 m/s when the driver saw an obstacle in the middle of the road 120 m ahead. The reaction time of the driver is 0.6 s. During this interval of 0.6 s, the bus was travelling at 20 m/s. After the brakes were applied, the bus travelled a distance of 100 m before stopping.

(a) Calculate the distance travelled by the bus during the reaction time of the driver.

(b) Calculate the deceleration of the bus.

(c) Explain whether the bus collided into the obstacle.

Solution:

(a) 12 m

(b) Deceleration = 2.0 m/s^2

(c) Total distance travelled = 112 m $<$ 120 m: not collided to the obstacle.